



Common problem types

In a previous video,

I shared how data analysis helped a company
figure out where to advertise its services.

An important part of this process
was strong problem-solving skills.

As a data analyst,
you'll find that problems are at the center
of what you do every single day,
but that's a good thing.

Think of problems as opportunities to put your skills to
work and find creative and insightful solutions.

Problems can be small or large,
simple or complex,
no problem is like another and they all require
a slightly different approach
but the first step is always the same:

Understanding what kind of problem you're trying to
solve and that's what we're going to talk about now.

Data analysts work with a variety of problems.

In this video, we're going to focus on six common types.

These include: making predictions, categorizing things,

spotting something unusual, identifying themes,
discovering connections, and finding patterns.

Let's define each of these now.

First, making predictions.

This problem type involves using data to make
an informed decision about
how things may be in the future.

For example, a hospital system might use
a remote patient monitoring to
predict health events for chronically ill patients.

The patients would take
their health vitals at home every day,
and that information combined with data about their age,
risk factors, and other important details could enable
the hospital's algorithm to predict
future health problems and
even reduce future hospitalizations.

The next problem type is categorizing things.

This means assigning information to
different groups or clusters based on common features.

An example of this problem type is
a manufacturer that reviews data on
shop floor employee performance.

An analyst may create a group for employees
who are most and least effective at engineering.

A group for employees who are most and least

effective at repair and maintenance,
most and least effective at assembly,
and many more groups or clusters.

Next, we have spotting something unusual.

In this problem type,

data analysts identify data

that is different from the norm.

An instance of spotting something
unusual in the real world is
a school system that has
a sudden increase in the number of students registered,
maybe as big as
a 30 percent jump in the number of students.

A data analyst might look into
this upswing and discover that
several new apartment complexes had been
built in the school district earlier that year.
They could use this analysis to make sure the school has
enough resources to handle the additional students.

Identifying themes is the next problem type.

Identifying themes takes categorization as a step
further by grouping information into broader concepts.

Going back to our manufacturer that has just
reviewed data on the shop floor employees.

First, these people are grouped by types and tasks.

But now a data analyst could

take those categories and group them into
the broader concept of
low productivity and high productivity.

This would make it possible for the business to
see who is most and least productive,
in order to reward top performers and
provide additional support to
those workers who need more training.

Now, the problem type of **discovering connections** enables
data analysts **to find**
similar challenges faced by different entities,
and then combine data and insights to address them.

Here's what I mean;
say a scooter company is experiencing
an issue with the wheels it gets from its wheel supplier.
That company would have to stop production until it could
get safe, quality wheels back in stock.
But meanwhile, the wheel companies encountering
the problem with the rubber it uses to make wheels,
turns out its rubber supplier could
not find the right materials either.
If all of these entities could talk about
the problems they're facing and share data openly,
they would find a lot of
similar challenges and better yet,
be able to collaborate to find a solution.

The final problem type is finding patterns.

Data analysts use data to find

patterns by using historical data to

understand what happened in

the past and is therefore likely to happen again.

Ecommerce companies use data to

find patterns all the time.

Data analysts look at transaction data to understand

customer buying habits at

certain points in time throughout the year.

They may find that customers buy more

canned goods right before a hurricane,

or they purchase fewer cold-weather accessories

like hats and gloves during warmer months.

The ecommerce companies can

use these insights to make sure

they stock the right amount of

products at these key times.

Alright, you've now learned six basic problem types

that data analysts typically face.

As a future data analyst,

this is going to be valuable knowledge for your career.

Coming up, we'll talk a bit more

about these problem types and I'll

provide even more examples of them

being solved by data analysts.

Personally, I love real-world examples.

They really help me better understand new concepts.

I can't wait to share

even more actual cases with you. See you there.